Pablo Valentín Cortés Castillo **Curriculum Vitae**

Research Interests & Background

My research develops a posteriori error control on parametric solution manifolds with Discontinuous Petrov–Galerkin (DPG) methods, seeking two-sided error bounds that hold for arbitrary trial functions, not only discrete solutions, and remain uniform along the manifold. The work spans the design of the DPG test-space topology, the construction of Fortin operators, and extensions to nonlinear and transport problems. Since the resulting estimators are computable, they can drive adaptivity, reduced-basis methods and PDE-constrained optimization. In particular, as variationally correct loss functions for learning parameter-to-solution maps with neural networks.

Interests: Reliable and efficient functionals and residuals, DPG methods, Fortin operators for broken spaces, parameter-to-solution manifolds and well-posedness of operators in Bochner spaces, non-uniform inf-sup conditions, operator learning, and high-performance simulation of physical phenomena.

Background: Finite element methods, PDEs, mathematical physics, inverse problems, PDE-constrained optimization and adjoint analysis, parallel algorithms.

Contact Information

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Education

Doctor of Philosophy in Mathematical Sciences - Candidate
Portland State University 2023 – June 2027 (expected)
Fariborz Maseeh Department of Mathematics and Statistics.
Advisor: Jay Gopalakrishnan.

Master of Science in Mathematical Sciences
Portland State University 2025
Fariborz Maseeh Department of Mathematics and Statistics.
Advisor: Jay Gopalakrishnan.

Master of Science in Engineering
Pontificia Universidad Católica de Chile 2024
Thesis: *Inverse problem of identification of Lamé parameters in elastodynamics equation using Symplectic HDG-FEM direct solver schemes.*
Advisor: Manuel Sánchez.

Professional Degree in Mathematical and Computational Engineering
Pontificia Universidad Católica de Chile 2024

Bachelor of Science in Engineering
Pontificia Universidad Católica de Chile 2021
- Major in Mathematical Engineering.
- Minor in Theory and Application of Mathematical Engineering.

Research Experience

Graduate Research Intern - Lawrence Livermore National Laboratory
Computational Engineering Division. 2026 – current

Supervisors: Mathias Schmidt & Boyan Lazarov.
Adjoint-based transient PDE-constrained optimization for elastodynamics in MFEM: topology optimization and inverse identification of material density.

Research Assistant - Portland State
Research assistantship grants at PSU Math + Stats Department.
Advisor: Jay Gopalakrishnan. Fall 2024 – Spring 2025, Winter 2026

Research

DPG loss functions for learning parameter-to-solution maps by neural networks 2026
Pablo Cortés Castillo, Wolfgang Dahmen and Jay Gopalakrishnan. Accepted in *Computers & Mathematics with Applications*. doi:10.1016/j.camwa.2026.09.005. ArXiv: https://arxiv.org/abs/2506.18773

Variationally correct loss functions for nonlinear parametric problems and their neural approximations 2026
Pablo Cortés Castillo, Wolfgang Dahmen and Jay Gopalakrishnan. Submitted to *IMA Journal of Numerical Analysis*. ArXiv: https://arxiv.org/abs/2610.19296

Global error estimators for parametric monotone nonlinearities and neural approximations 2026
Pablo Cortés Castillo, Wolfgang Dahmen and Jay Gopalakrishnan. Submitted to *Advances in Computational Mathematics*. ArXiv: https://arxiv.org/abs/2610.05767

Fortin operators for DPG advection discretizations 2026
Pablo Cortés Castillo and Jay Gopalakrishnan. Submitted to *Mathematics of Computation*. ArXiv: https://arxiv.org/abs/2608.30043

Inverse problem of identification of Lamé parameters in elastodynamics equation using Symplectic HDG-FEM direct solver schemes. 2024
Pablo Cortés Castillo. Master Thesis.

Selected Talks

MFEM Community Workshop September 2026
Talk: *Transient PDE-constrained optimization for hyperbolic phenomena*. Georgia Tech. Atlanta, GA.

SIAM PNW Biennial Meeting October 2025
Mini-symposium: *Recent Developments on Hybrid Methods Combining Neural Networks with Classical Numerical Methods*. University of Washington. Seattle, WA.

Leadership & Service

Society of Industrial and Applied Mathematics (SIAM) 2025 – current
Membership.

Research Training Group in Computation and Data Enabled Science (RTG in CADES) 2023 – current
Graduate Trainee.

Skills

Programming: Python (Advanced), C++ (Advanced), PyTorch, Matlab, Mathematica, R, SQL, HPC, NGSolve, MFEM.
Languages: Spanish (native), English (proficient).

Awards & Funding

SIAM Travel Award 2025 October 2025
SIAM PNW Biennial Meeting travel award. University of Washington. Seattle, WA.

Frank Cater Scholarship 2023 – 2025
Awarded by the Fariborz Maseeh Department of Mathematics and Statistics. Portland State University. Portland, OR.

Teaching Experience

Lecturer - Portland State
Lecturer at PSU Math + Stats Department.

- **MTH111Z - Precalculus** Fall 2026
- **MTH111Z - Precalculus** Fall 2025
- **Bootcamp - Introduction to Finite Element Methods** Summer 2025
- **Bootcamp - Introduction to Finite Element Methods** Summer 2024

Teaching Assistant - Portland State
TA and grader for undergraduate courses at PSU Math + Stats Department.

- **Intro. to Linear Algebra** Spring 2024
- **Calculus II** Winter 2024
- **Trigonometry** Fall 2023

Teaching Assistant - PUC Chile
TA and grader for graduate and undergraduate courses at the PUC Engineering and Mathematics Department.

- **Advanced Linear Algebra and Applications** Fall 2022
- **Advanced Linear Algebra and Applications** Spring 2022
- **Precalculus** Fall 2021
- **Linear Algebra** Fall 2019
- **Calculus** Spring 2019
- **Statics & Dynamics** Fall 2018

Industry Experience

Servicios Financieros Lider-BCI. 2022 – 2026
Data & Analytics.
Design and Implementation of different predictive models to accurately forecast clients transactional behavior (for example churn, loans, etc.), using statistics and machine learning techniques. Construction of pipelines, architecture developing and maintenance among other tasks are also part of the job.